

Distilling Native Query–Key Responses into Compact Memory Landmarks

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Abstract

Progressive Retrieval Attention (PRA) uses compact routing indexes to choose long-memory chunks, then materializes the selected chunks’ unchanged native key/value (K/V) states. We ask whether low-rank native query–key interactions recover address signals erased by one mean gist. An initial gated study and a 120-controller sweep fail a prespecified response-distillation gate, but expose an evidence-supervised rank-16 QASPER frontier. We then freeze that configuration and evaluate 64 fresh HotpotQA and 64 fresh QASPER identities with five seeds and a fixed four-chunk budget. On QASPER, rank-16 recall is 0.258 versus 0.117 for a native mean (paired +0.141, 95% interval [0.080, 0.210]); on HotpotQA it is 0.120 versus 0.125, while BM25 reaches 0.386. A Llama-family SmolLM2 replication preserves the QASPER-over-mean effect (+0.064, interval [0.019, 0.115]), but exact routing remains stronger. Native-K/V generation separates routing from memory use: rank-16 improves Hotpot gold-token log-probability by 0.074, yet on QASPER it is -0.212 versus no memory while remaining +0.312 above shuffled memory. Better routing therefore mitigates wrong memory without solving frozen late-layer integration. A vectorized index stores rank-16 projected tokens in FP16 or INT8, supports stable batched top- k and append updates, and leaves backing native K/V unchanged. Compact native interaction subspaces are a portable QASPER routing channel, not a universal response compressor or a complete memory-use solution.

1 Introduction

PRA separates *routing* from *reading*. A small summary chooses memory chunks; only the chosen chunks contribute their original token K/V to attention. The separation preserves pretrained values and bounds active memory, but it makes routing-summary quality consequential. One mean vector per chunk is attractive because it is deterministic, cacheable, and cheap. It is also a severe compression: a sparse key that responds strongly to a future query can disappear inside 31 unrelated tokens.

Earlier work already tested natural repairs. Multi-gist, prototype, k -means, SOM, local-gist, native-QK closure, lexical, and graph-query variants sometimes helped a dataset or controlled fixture, but did not establish a uniform natural frontier. Those methods mainly organize semantic or local geometry. This paper asks a different question: can a compact native-key set preserve the *response surface* that the frozen attention layer assigns to the full chunk?

The distinction matters. A key is useful because of how plausible queries address it, not because it reconstructs the mean of its neighbors or reads well as text. We therefore use full token-level keys as a teacher, measure ranking preservation directly, and keep the downstream materialization budget fixed. The study is gated. We first verify inherited Paper 2.6 rows, then test whether the teacher is meaningful, then ask whether a query-aware oracle has headroom. A learned selector is permitted only after those checks; more expressive off-manifold slots and recurrent memory remain closed if it fails.

The contributions are:

1. a tensor-native definition of chunk response distillation under grouped-query attention, with max, top- r , log-sum-exp, and attention-mass teachers;
2. matched native-key controllers, response metrics, routing metrics, changed-selection audits, and synchronized CUDA cost measurements;
3. a controlled 32–256-token suite and a frozen natural evaluation aligned with the Paper 2.5/2.6 32-token chunk and four-chunk budget;
4. a measured boundary: oracle landmarks show natural QASPER headroom, while a 321-parameter key-only selector recovers too little of that headroom to pass deployment gates;
5. a 120-controller query-conditioned sweep separating a prespecified primary result from test-selected exploratory findings and showing that response preservation, evidence ranking, and dataset utility are not interchangeable;
6. a post-G3 diagnostic frontier that crosses matched native geometry with direct rank-8/16/32 native-key projections, four/eight static low-rank representatives, and measured routing-index bytes and latency without changing native-K/V materialization.
7. a 128-identity confirmation with native-K/V generation and causal controls, a Llama-family/tokenizer replication, and a quantized vectorized serving index.

2 Native response distillation

2.1 Full-key teacher

Let chunk C contain pre-RoPE native keys $K_C \in \mathbb{R}^{n \times H_{kv} \times d_h}$ and let a frozen decoder produce query heads $q \in \mathbb{R}^{H_q \times d_h}$. The grouped-query map $g(h)$ assigns each query head to its native key head. Scaled token responses are

$$a_{i,h}(q, C) = \frac{q_h^\top k_{i,g(h)}}{\sqrt{d_h}}. \quad (1)$$

The teacher reduces tokens per head and then averages heads:

$$S^*(q, C) = \frac{1}{H_q} \sum_h F(a_{1,h}, \dots, a_{n,h}). \quad (2)$$

We validate four reductions: maximum, top- r mean, length-normalized log-sum-exp, and a softmax attention-mass proxy. Validation selects top-four mean with $r = 4$. This selection occurs before test evaluation and is frozen for every controller.

For compact landmarks $G(C) = \{\hat{k}_1, \dots, \hat{k}_m\}$, the estimate $\hat{S}(q, C)$ applies the same F and grouped-head mapping. The primary target is not $\|K_C - \hat{K}_C\|$. It is agreement between S^* and $\hat{S}$ over candidate chunks: score error, rank correlation, teacher top- k overlap, and KL divergence between softmax-normalized chunk distributions.

2.2 Controllers

All deployable landmark outputs are actual pretrained keys. Mean is the only synthetic control.

Mean Average valid keys independently in each native key head, yielding one key per chunk.

Last and random

Retain the final m valid keys or a deterministic random subset.

Farthest-first

Flatten native heads per token and greedily retain cosine-diverse keys.

Greedy oracle

Add the token that most reduces full-response squared error. It sees the evaluation query and is explicitly non-deployable.

Learned selector

Score each token from overall key norm, mean head norm, head-norm variation, cosine to the chunk centroid, local key change, forward/reverse position, and validity. A two-layer MLP, $8 \rightarrow 32 \rightarrow 1$, has 321 parameters. Hard top- m is used at inference.

Oracle selections for $m = 8$ supervise the learned model. Five seeds train for 400 AdamW steps on 16 validation identities. The backbone, native projections, tokenizer, and all natural test identities remain frozen. The selector receives no question vector, evidence label, dataset identifier, lexical feature, or chunk-level answer supervision at inference.

2.3 Query-conditioned continuation

The continuation addresses the key-only selector’s missing information need without changing native keys. Let $f_i \in \mathbb{R}^8$ be token i ’s normalized key-only features and let $\bar{q} \in \mathbb{R}^{H_q d_h}$ be the flattened, RMS-scaled native query. The selector logit is

$$z_i = \text{MLP}(f_i) + \frac{(W_q \bar{q})^\top (W_f f_i)}{\sqrt{r}}, \quad (3)$$

where $r \in \{8, 16, 32\}$. The resulting models have 16,769, 33,217, and 66,113 parameters. Hard top- m selects actual native keys at inference.

Training uses a piecewise-differentiable surrogate only to move selector logits. Within each chunk, the hard top- m support is retained while a softmax over its logits weights scalar native token responses. This gives a chunk surrogate $\tilde{S}(q, C)$. Final evaluation discards the surrogate, gathers hard-selected keys, and recomputes $\hat{S}(q, C)$ with the exact grouped-query native-QK function.

Four objectives are prespecified. *Oracle imitation* applies token-level weighted binary cross-entropy to greedy response-preserving selections. *Listwise* minimizes cross-entropy from the chunk softmax to a uniform distribution over evidence chunks. *Combined* adds $\text{KL}(\text{softmax } S^* \| \text{softmax } \tilde{S})$ to listwise ranking. *Decision-aware* adds pairwise softplus penalties between evidence chunks and non-evidence chunks nearest the fourth-place routing threshold. We cross all four objectives with both landmark counts, all three ranks, and five seeds: 120 controllers, each trained for 300 AdamW steps on validation only. Combined $r = 16, m = 4$ is the prespecified primary configuration; every other test comparison is reported without replacing it.

2.4 Post-G3 native geometry and low-rank routing

The original preregistration stops at G3. A separate diagnostic extension asks whether simple multi-modal geometry explains the landmark result and whether routing needs full native-key width. First, deterministic Euclidean k -means centroids, cluster medoids, and farthest-first actual keys are constructed inside every 32-token chunk for $m \in \{1, 2, 4, 8\}$. Centroids are routing-only; selected chunks still materialize their unchanged token K/V. These controls use the same top-four-mean response reduction as the frozen teacher.

For direct low-rank routing, flatten the eight native key heads into $u_i \in \mathbb{R}^{1024}$ and the 16 native query heads into $\bar{q} \in \mathbb{R}^{2048}$. A validation-derived scalar RMS normalizes keys without changing relative

token norms. The learned score is

$$\ell_i = \frac{(W_q \bar{q})^\top (W_k u_i)}{\sqrt{r}}, \quad W_q \in \mathbb{R}^{r \times 2048}, \quad W_k \in \mathbb{R}^{r \times 1024}, \quad (4)$$

for $r \in \{8, 16, 32\}$. The models contain 24,576, 49,152, and 98,304 parameters. Combined listwise evidence ranking and native-response distillation train the projections for 300 steps on validation identities. The backbone, native projections, and test identities remain frozen.

At index-construction time, $z_i = W_k u_i$ is cached. Online work is $z_q = W_q \bar{q}$ followed by rank- r dots; no deployable low-rank row scans full native keys. We evaluate all 32 projected tokens and static k -means, medoid, and farthest-first indexes with $m \in \{4, 8\}$. Thus token compression ($32 \rightarrow m$) and interaction compression ($1024 \rightarrow r$) are measured separately and jointly. With FP32 routing indexes, a chunk uses 131,072 bytes for 32×1024 full native keys, 4,096 for one mean key, $4mr$ for static low-rank representatives, and $128r$ for all projected tokens.

This accounting separates routing-index residency from backing native K/V, selected-K/V transfer, and active materialization. Query-conditioned native landmark selection also needs its full token selector index because landmark identities vary with the query; it cannot honestly be counted as only m persistent native keys. Construction time, cached online latency, native dots, low-rank dots, and eager peak-GPU deltas are recorded independently.

2.5 Matched materialization

The intervention changes only routing summaries. Candidate units are the same 32-token local chunks. Every method requests four chunks, so it materializes approximately 128 native K/V tokens: 126.1 on HotpotQA because of short final chunks and 128.0 on QASPER. Active memory fractions are 8.8% and 3.3%, respectively. Values are never compressed, projected, or synthesized.

3 Protocol

3.1 Frozen natural cohort

We reuse Qwen3-0.6B revision `c1899de2`, layer 27, 16 query heads, eight K/V heads, and head width 128. Source blocks are encoded in bounded 256-token calls, then divided into 32-token routing chunks without local re-encoding. Pre-RoPE keys and attention-input query states preserve token spans and evidence masks. The query state is projected through the unchanged native `q_proj` and `q_norm`.

Validation contains eight HotpotQA and eight QASPER identities. Test contains 16 of each, matching the frozen Paper 2.5/2.6 identities and excluding validation identities. Feature tensors are 329 MB and 621 MB, respectively; manifests record SHA-256 hashes. Paper 2.6’s 224 test rows are re-aggregated exactly at G0. Paper 2.5’s 256-token-parent, 20%-budget rows are retained only as historical context because their candidate unit and budget are not numerically matched.

3.2 Controlled suite

Twenty synthetic cases cross chunk sizes $n \in \{32, 64, 128, 256\}$. Each case has 16 candidate chunks. An evidence chunk contains sparse query-aligned native keys while a distractor has a coherent but weaker mean. This tests whether full-key and landmark controllers preserve a sparse address that averaging can hide. The suite is a mechanism check, not evidence for natural language performance.

3.3 Metrics and uncertainty

Native preservation metrics are MAE, RMSE, Spearman ρ , Kendall τ , KL divergence, and teacher top- k overlap for $k \in \{1, 2, 4, 8\}$. Routing metrics are evidence recall and precision, any-evidence recall, contiguous evidence-group completion, exact selected identity, MRR, active memory fraction, evidence-normalized overhead, materialized tokens, native dots, and wall time. Five-thousand-draw paired bootstrap intervals resample identities. Selector seeds are averaged within identity before comparisons with deterministic baselines.

The prespecified gates are:

- G0: inherited Paper 2.6 aggregation parity;
- G1: full-K beats mean on controlled recall and at least one natural dataset;
- G2: greedy $m \leq 8$ improves teacher top-four overlap and at least one natural recall result without changing materialized K/V;
- G3: the learned selector recovers at least 80% of oracle gain, improves recall or completion on one natural dataset, and regresses by less than 0.02 elsewhere;
- G4/G5: synthetic slots and streaming memory run only after G3 passes.

The post-G3 extension uses E0–E6 and does not revise G3. E0 verifies frozen artifact, cache, model, and identity parity. E1 compares query/response-aware selection with matched native geometry. E2 applies a cross-dataset non-regression rule to query conditioning; test-selected rows remain exploratory. E3 compares ranking/decision objectives with oracle imitation. E4 requires at least 95% retention of the best full-dimensional retrieval gain on a natural dataset and at least $4\times$ less routing state. E5 asks whether joint m, r compression matches the QASPER semantic gist on a better quality–bytes–latency point. E6 opens 2Wiki/MuSiQue only if E2–E5 form a credible frontier.

4 Results

4.1 Teacher sanity and oracle headroom

G0 passes with zero mismatches across 224 inherited rows. Validation selects top-four mean across query heads: its validation evidence recall is 0.1387 versus 0.1255 for the corresponding mean-key control. Max, log-sum-exp, and attention-mass teachers do not beat that control on validation.

G1 also passes, narrowly on natural data. Full-K recall is 1.000 versus 0.950 for mean in the controlled suite. On held-out HotpotQA it is 0.1240 versus 0.1235; on QASPER it is 0.1252 versus 0.0976. The natural teacher is therefore not a strong retrieval oracle by itself. It provides a modest QASPER signal and essentially no Hotpot gain.

G2 passes. Relative to mean, greedy $m = 8$ improves aggregate teacher top-four overlap by 0.3438. Recall is not monotone in m : $m = 4$ is best, improving HotpotQA by 0.0219 and QASPER by 0.0802. The paired intervals, $[-0.0313, 0.0750]$ and $[-0.0142, 0.2233]$, both include zero. Oracle landmarks therefore establish response-preservation headroom and a promising QASPER direction, not a statistically resolved natural retrieval gain.

4.2 Learned selector: useful direction, failed gate

The learned selector does not recover the oracle. At $m = 8$, it captures 40.5% of the oracle teacher-overlap gain, below the 80% threshold. Its HotpotQA recall delta over the QK mean is +0.0004 with

Table 1: Held-out four-chunk routing. Learned rows average five seeds. Oracle rows use the test query and are non-deployable. Values are evidence recall; parentheses give teacher top-four overlap.

Method	HotpotQA	QASPER
Mean key, $m = 1$	0.124 (0.594)	0.098 (0.516)
Full-K teacher	0.124 (1.000)	0.125 (1.000)
Farthest-first, $m = 2$	0.142 (0.734)	0.130 (0.422)
Greedy oracle, $m = 4$	0.145 (0.922)	0.178 (0.703)
Greedy oracle, $m = 8$	0.124 (0.984)	0.136 (0.812)
Learned selector, $m = 4$	0.122 (0.741)	0.083 (0.422)
Learned selector, $m = 8$	0.124 (0.834)	0.139 (0.553)

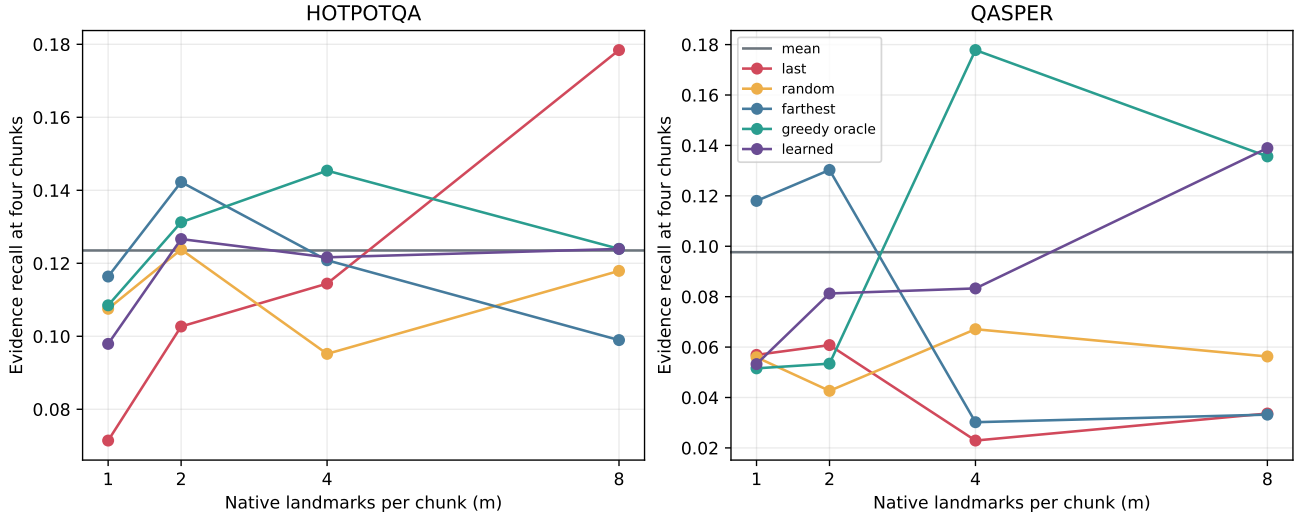


Figure 1: Natural evidence recall under the matched four-chunk materialization budget. The dataset-separated view exposes the Hotpot/QASPER asymmetry.

interval $[-0.0573, 0.0533]$. Its QASPER delta is $+0.0413$ with interval $[-0.0589, 0.1670]$. G3 therefore fails and G4/G5 do not run.

Seed variation is material. HotpotQA $m = 8$ recall ranges from 0.1173 to 0.1298. QASPER ranges from 0.0478 to 0.1821; four seeds are at or above 0.119, while seed 37 collapses. Teacher top-four overlap is more stable than evidence recall, showing that response imitation and evidence utility are related but not interchangeable.

4.3 Query conditioning: objective-dependent utility

The prespecified combined $r = 16, m = 4$ controller improves HotpotQA recall from the full-key teacher’s 0.1235 to 0.1667, a paired difference of 0.0432 with interval $[-0.0112, 0.0970]$. It also improves over the key-only $m = 8$ selector by 0.0427, interval $[-0.0015, 0.0907]$. This is the best HotpotQA result in the 24-configuration continuation, but remains below BM25, exact, and hybrid routing. The same controller reaches only 0.0574 on QASPER, 0.0402 below the mean-key control and 0.1202 below exact routing. Query conditioning therefore does not produce a dataset-uniform primary result.

The test-selected exploratory result is more encouraging but is not confirmatory. Decision-aware $r = 32, m = 8$ reaches 0.1718 QASPER recall, compared with 0.1776 for Paper 2.6 exact routing, 0.1389 for the key-only selector, and 0.0976 for the mean key. Its paired differences are -0.0059 against exact,

Table 2: Matched four-chunk evidence recall. “Primary” was fixed before test evaluation. “Exploratory” is the best QASPER row selected after inspecting the 24-configuration test matrix.

Method	HotpotQA	QASPER
QK mean	0.124	0.098
Key-only learned, $m = 8$	0.124	0.139
Paper 2.6 BM25	0.398	0.042
Paper 2.6 exact	0.287	0.178
Paper 2.6 hybrid	0.279	0.081
Primary combined, $r = 16, m = 4$	0.167	0.057
Exploratory decision-aware, $r = 32, m = 8$	0.143	0.172

interval $[-0.1831, 0.2049]$, and $+0.0328$ against key-only, interval $[-0.0292, 0.1073]$. Five seed means span 0.1639–0.1796. The narrow seed range shows optimization stability for this configuration; the wide identity-bootstrap intervals show that 16 test questions per dataset are still too few to resolve comparative retrieval utility.

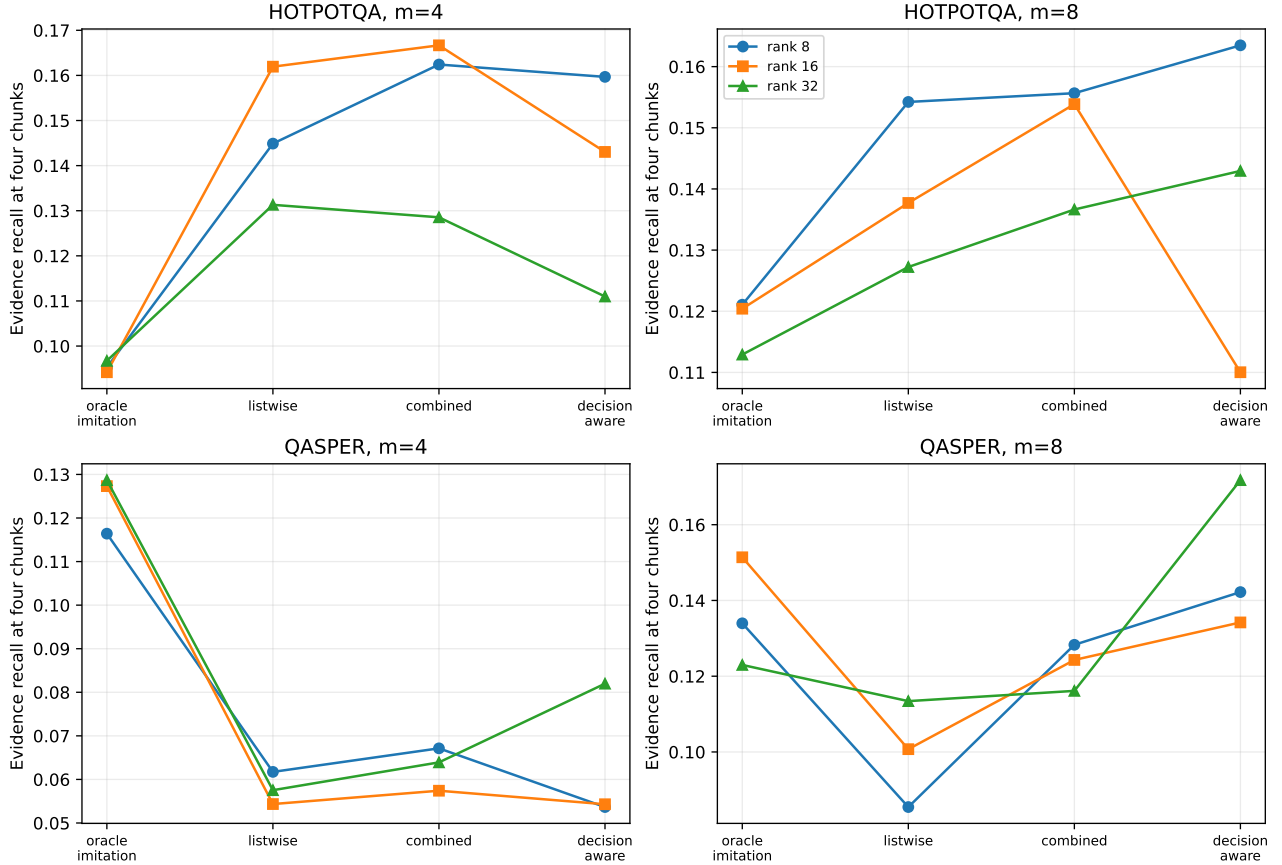


Figure 2: Evidence recall across objective, low-rank width, and landmark count. Objective choice changes both the level and the dataset direction; additional rank or landmarks are not monotonically beneficial. Points average five selector seeds.

Response preservation does not select the same controller as retrieval utility. Oracle-imitation $r = 32, m = 8$ gives the best aggregate teacher top-four overlap, 0.717 versus 0.555 for mean and

0.898 for greedy oracle, recovering 47.3% of oracle gain. Yet across the 24 configurations, teacher overlap and evidence recall correlate negatively on HotpotQA ($r = -0.665$) and positively on QASPER ($r = 0.682$). These descriptive correlations do not establish causality, but they make the failure mode visible: matching a frozen layer’s response can preserve preferences that are not evidence-bearing.

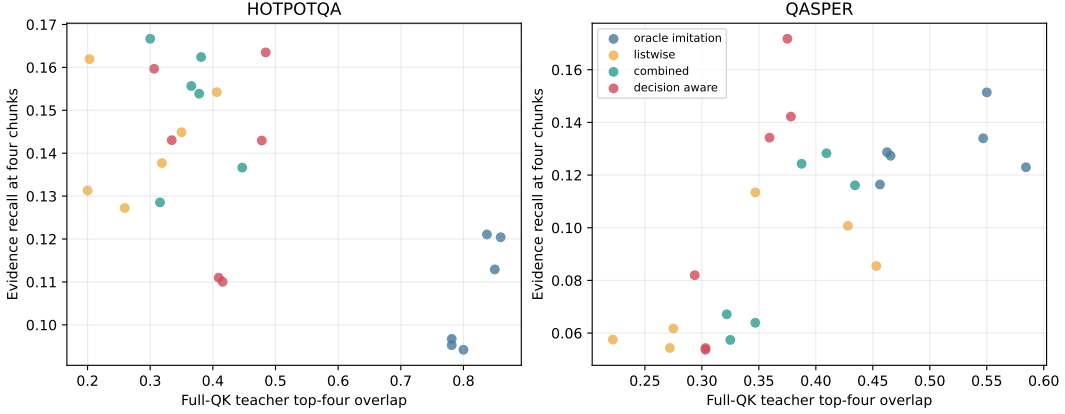


Figure 3: Response preservation versus evidence recall for all 24 query-conditioned configurations. The opposing dataset trends explain why the highest-fidelity compressor is not a universal retrieval controller.

The primary selector has 33,217 parameters and trains in 36.2 seconds per seed; the exploratory selector has 66,113 parameters and trains in 50.2 seconds per seed on the GTX 950M. Synchronized selector-plus-top- m latency is 42.6/117.0 ms for the primary and 42.9/120.5 ms for the exploratory controller on HotpotQA/QASPER. These continuation timings exclude compact-key QK scoring and materialization, so they are component measurements rather than direct replacements for Table 7.

4.4 Post-G3 low-rank frontier

Simple native geometry does not explain the continuation result. At matched $m = 2$, farthest-first is the strongest geometric control on both datasets, reaching 0.1423 HotpotQA and 0.1303 QASPER recall. Increasing to four or eight centroids, medoids, or farthest points is not monotone. In particular, four native centroids reach only 0.0332 on QASPER despite 0.453 teacher top-four overlap. Preserving several modes of native-key geometry is therefore insufficient by itself.

Direct low-rank native interaction gives a different result. Scoring all 32 projected tokens at rank 16 reaches 0.2542 QASPER recall, a gain of 0.1566 over the native mean with paired interval $[0.0462, 0.3001]$. It also exceeds the test-selected query-conditioned controller by 0.0825, interval $[0.0366, 0.1337]$. The gain over Paper 2.6 exact routing is 0.0766, but its interval $[-0.0791, 0.2763]$ includes zero. Five seed means are 0.1940–0.3027. Rank 8 is close at 0.2456; rank 32 falls to 0.2179 and has the largest seed standard deviation, 0.0666. Extra rank is not reliably useful.

The all-token rank-16 index is $64\times$ smaller than the 32×1024 full native-key stream and $2\times$ smaller than one full-width mean key. Joint token/rank compression is more aggressive: rank-8 k -means with eight representatives uses 256 bytes per chunk, $512\times$ below the full stream and $16\times$ below the mean, while reaching 0.1829 QASPER recall. Its gain over exact is only 0.0052 with interval $[-0.1250, 0.1675]$. Index construction is expensive in this reference implementation (2.26 seconds per QASPER example for rank-8 k -means), but it is offline and cacheable; cached online scoring is 0.96 ms. These timings are component measurements on the GTX 950M, not production throughput claims.

Table 3: Post-G3 diagnostic frontier. Bytes are persistent FP32 routing-index bytes per 32-token chunk; cached latency excludes index construction and native-K/V materialization. Low-rank rows average five seeds.

Method	Bytes/chunk	Cached ms	HotpotQA	QASPER
Native mean, $d = 1024$	4,096	1.11/1.19	0.124	0.098
Native farthest, $m = 2$	8,192	2.76/5.27	0.142	0.130
Low-rank all, $r = 8$	1,024	0.86/0.88	0.104	0.246
Low-rank all, $r = 16$	2,048	0.90/0.96	0.121	0.254
Low-rank all, $r = 32$	4,096	0.90/0.92	0.145	0.218
Low-rank k -means, $r = 8, m = 8$	256	0.90/0.96	0.100	0.183
Paper 2.6 exact	—	—	0.287	0.178

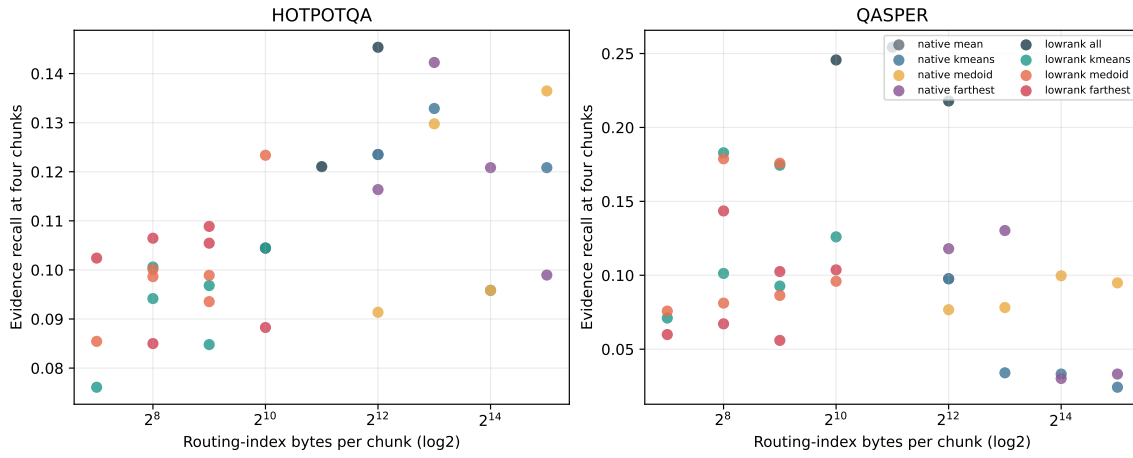


Figure 4: Evidence recall against persistent routing-index bytes per chunk. Direct low-rank routing creates a new QASPER quality-memory frontier; Hotpot remains dominated by lexical routing.

Low teacher overlap limits the interpretation. Rank-16 all-token routing overlaps the teacher’s top four by only 0.228 on QASPER and 0.309 on HotpotQA. The selected chunks differ from the mean-key selection for every test identity. This controller has learned an evidence-bearing interaction subspace from validation labels; it has not distilled the frozen teacher faithfully.

Structural ablations reinforce the dataset asymmetry. On HotpotQA, query-independent salience at $m = 4$ reaches 0.1813, above every bilinear variant in the ablation. On QASPER, bilinear-only rank 32, $m = 8$ reaches 0.1488, above its salience-plus-bilinear counterpart at 0.1161. These are test-matrix diagnostics, not prespecified winners. They show that query interaction contributes on QASPER while learned token importance can suffice on Hotpot; simply adding both terms does not guarantee a better selector.

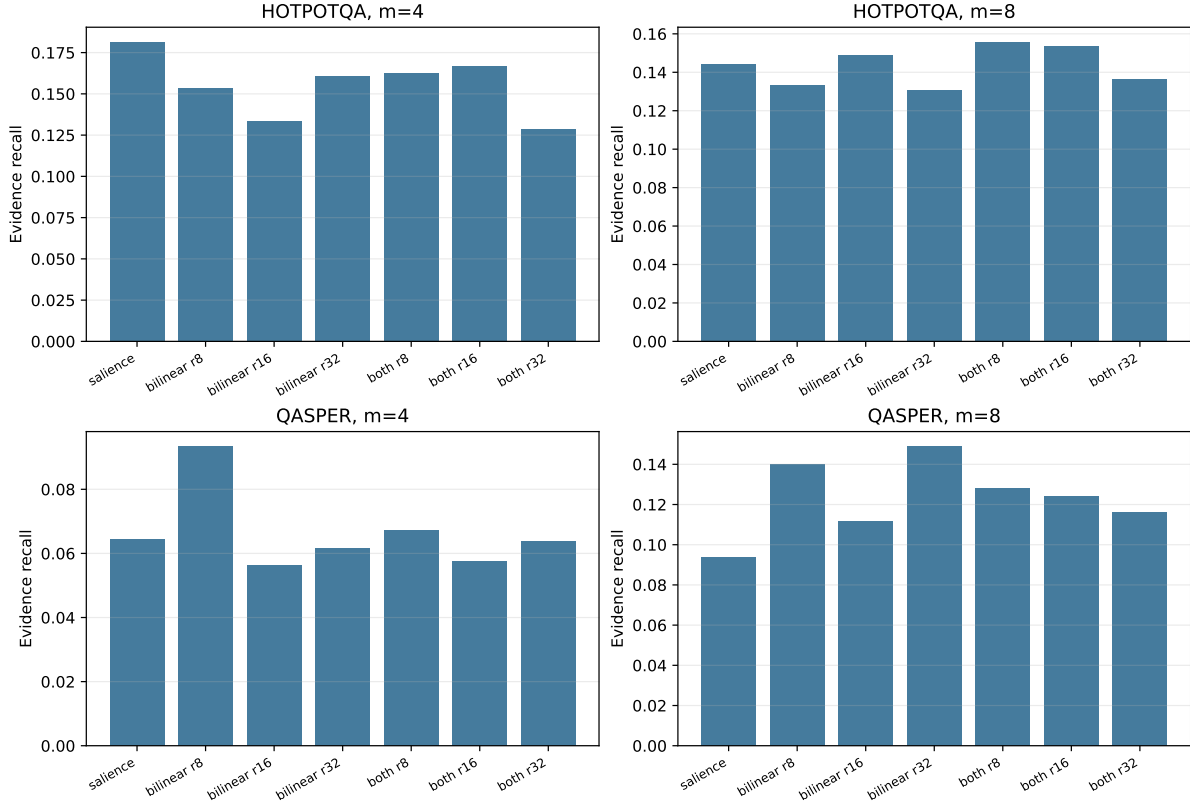


Figure 5: Selector structure ablation at four and eight landmarks. Saliency and bilinear query interaction have different dataset roles, and their combination is not uniformly additive. Points average five seeds.

4.5 Fresh confirmation and causal memory use

We froze rank 16, rank-8/eight-centroid construction, five checkpoints, chunk size, and four-chunk budget before opening a new offset-24 cohort of 64 identities per dataset. No confirmation identity overlaps the earlier natural matrix. Table 4 shows that the low-rank effect is reproducible but dataset-specific. Rank 16 improves QASPER recall over mean by 0.1407, interval $[0.0795, 0.2099]$, and over hybrid by 0.1149, interval $[0.0562, 0.1796]$. Its -0.0600 difference from exact remains unresolved. Neither low-rank index improves HotpotQA over mean, and all remain far below lexical channels.

Table 4: Fresh 64+64 confirmation, evidence recall at four chunks. Learned rows ensemble five validation-trained seeds.

Method	HotpotQA	QASPER
Native mean	0.125	0.117
BM25	0.386	0.065
Exact	0.282	0.318
Hybrid	0.224	0.143
Rank 16, all projected tokens	0.120	0.258
Rank 8, eight centroids	0.129	0.206
Evidence oracle	0.814	0.731

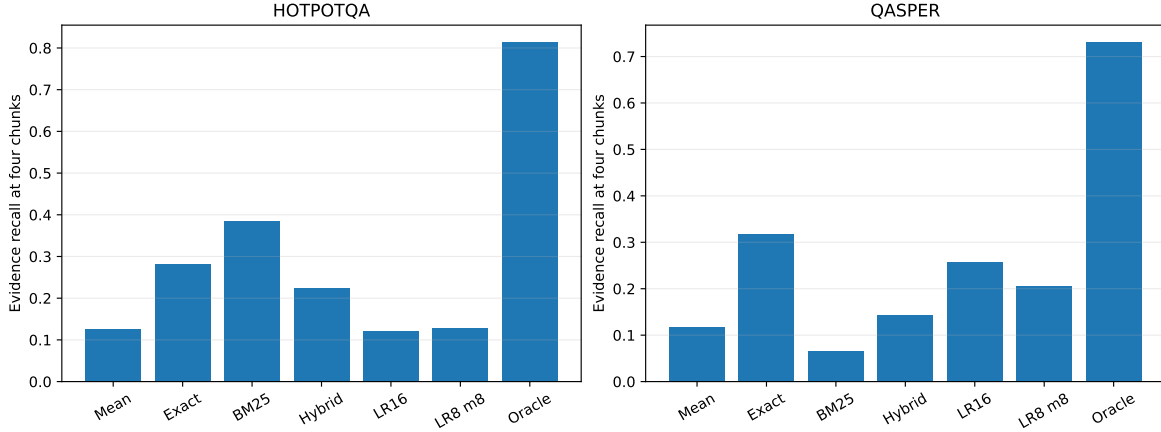


Figure 6: Fresh-cohort routing confirmation. The rank-16 gain transfers within QASPER, while Hotpot remains lexical.

We next replay the saved chunk identities through unchanged layer-27 native K/V. Teacher-forced gold-token log-probability is primary; 12-token greedy EM/F1 is secondary because the 0.6B frozen model rarely emits exact answers. Table 5 separates absolute memory use from routing quality. On HotpotQA, rank 16 improves over no memory by 0.0738, interval $[0.0074, 0.1400]$, but differs from shuffled memory by only 0.0091, with an interval crossing zero. On QASPER, rank 16 is -0.2124 versus no memory, interval $[-0.4079, -0.0194]$, yet exceeds shuffled memory by 0.3115, interval $[0.1378, 0.4887]$, and bottom-ranked memory by 0.4699, interval $[0.2369, 0.7118]$. Selection is causally better than wrong memory, but the frozen late-layer consumer does not turn the retrieval gain into a net benefit.

Table 5: Mean gold-token log-probability change relative to no memory. All memory rows request four chunks.

Condition	HotpotQA	QASPER
Native mean	+0.087	-0.095
Rank 16	+0.074	-0.212
Rank 8, eight centroids	+0.034	-0.194
Shuffled chunks	+0.065	-0.524
Bottom-ranked chunks	-0.035	-0.682
Evidence oracle	+0.076	-0.092
Direct evidence text	+2.956	+0.225

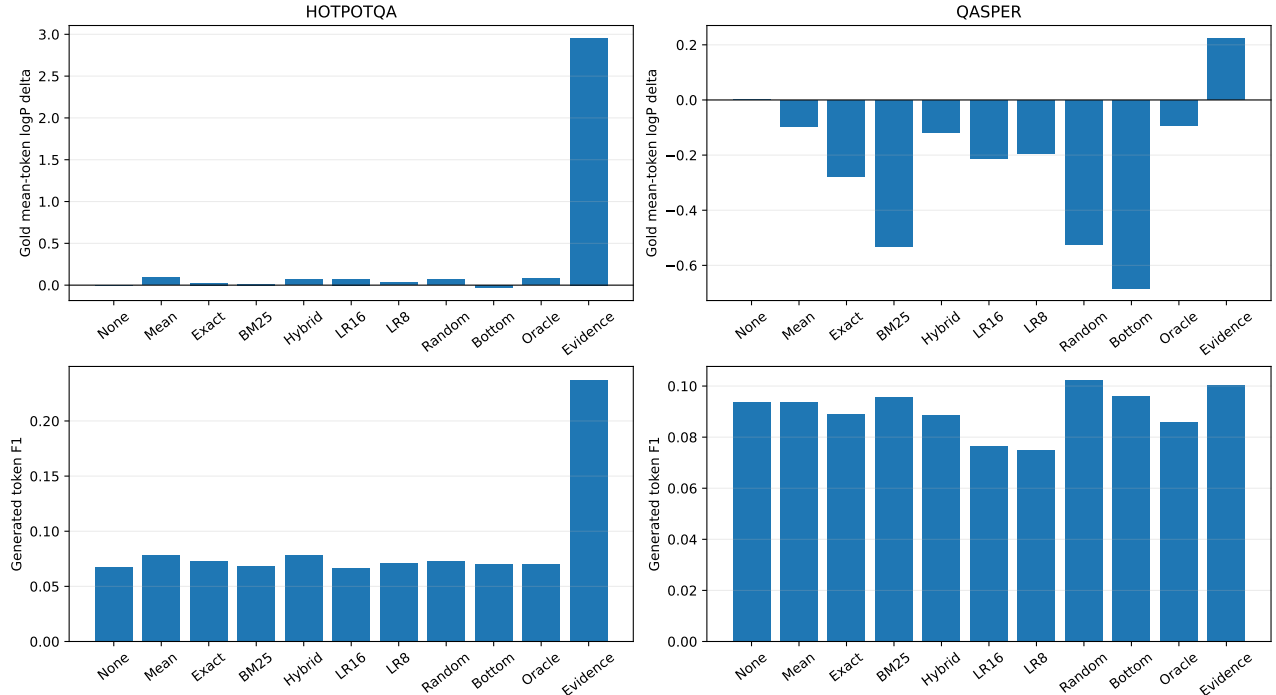


Figure 7: Native-K/V generation and causal controls. Routing quality and memory integration are distinct bottlenecks.

4.6 Cross-model replication

We repeat validation training and fresh-cohort routing on SmolLM2-135M, a Llama-family model with a different tokenizer, 30 layers, 9 query heads, 3 native K/V heads, and 64-wide heads. The capture uses layer 29 and the model’s own normalized pre-RoPE query/key projections. Rank-16 QASPER recall is 0.155 versus 0.092 for mean, a paired gain of 0.0638 with interval [0.0192, 0.1148]. Rank-8/eight-centroid recall is 0.165, gain 0.0739 with interval [0.0226, 0.1342]. Exact reaches 0.298 and remains significantly stronger. On HotpotQA, neither learned index improves over mean and BM25 again leads. The low-rank QASPER signal is therefore portable across model family and tokenizer, but its channel ordering is also portable: exact lexical evidence remains the stronger control.

Table 6: SmolLM2/Llama-family fresh-cohort recall.

Method	HotpotQA	QASPER
Native mean	0.159	0.092
BM25	0.394	0.059
Exact	0.277	0.298
Hybrid	0.246	0.117
Rank 16	0.132	0.155
Rank 8, eight centroids	0.135	0.165

4.7 Production index

The reference serving index caches projected tokens as FP32, FP16, BF16, or symmetric per-rank INT8. It scores all chunks for a query batch in one tensor operation, applies stable top- k , accounts for masks and quantization scales, and appends new chunks without rebuilding old rows. Eight-centroid modes

perform deterministic clustering offline. Backing native K/V is unchanged and is materialized only after routing.

On the GTX 950M, rank-16 FP16 occupies 1,056 bytes per 32-token chunk including its mask, versus 2,080 bytes for FP32. INT8 uses about 545 bytes; INT8 with eight centroids uses about 137 bytes. Warm single-query search is 1.5–2.1 ms and an eight-query batch sustains roughly 2,660–4,090 queries/s. Kernel-launch overhead dominates at this cohort size, so quantization primarily reduces resident index memory rather than latency. FP16 and INT8 preserve the seed-11 FP32 selections exactly in aggregate (0.118/0.230 Hotpot/QASPER recall); centroid modes trade recall for residency. Deterministic centroid construction remains expensive offline: roughly 0.7 s per Hotpot example and 2.0 s per QASPER example. Append updates average below 0.21 ms in every mode except one warm-up-tainted Hotpot FP32 mean. These are component measurements, not end-to-end serving throughput.

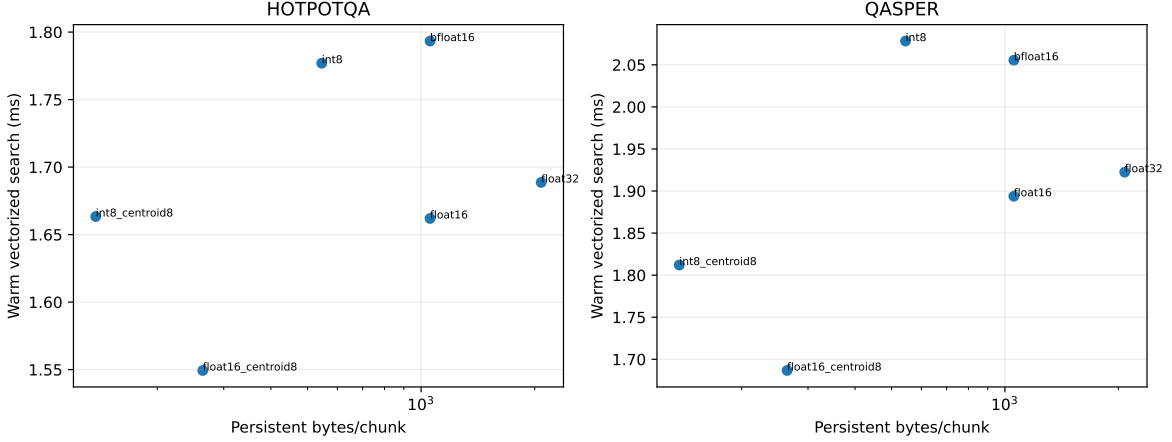


Figure 8: Warm vectorized search versus persistent routing-index bytes. Quantization changes residency far more than latency on this small, old GPU.

4.8 Comparison with Paper 2.6

Figure 9 compares the same frozen identities and four-chunk budget. Learned QK $m = 8$ is not a new overall frontier. On HotpotQA its 0.124 recall is below Paper 2.6 BM25 (0.398), exact (0.287), and iterative hybrid (0.279), with paired intervals excluding zero against all three. On QASPER it reaches 0.139: above Paper 2.6’s semantic gist (0.038) by 0.1010, with paired interval $[0.00004, 0.2270]$; above iterative hybrid (0.081) by 0.0579, interval $[-0.0610, 0.1974]$; and below exact routing (0.178) by 0.0387, interval $[-0.1951, 0.1334]$. The positive, defensible claim is dataset-specific: native response distillation improves the old QASPER semantic-gist baseline, but does not displace lexical exact matching and does not transfer to Hotpot.

Table 7: End-to-end routing-controller cost at $m = 8$. Mean uses $m = 1$. The full-K row performs no selection.

Method	HotpotQA (50.1 chunks)		QASPER (142.6 chunks)	
	ms	native dots	ms	native dots
Mean	6.1	802	10.7	2,282
Full-K	25.9	24,898	34.1	70,853
Farthest-first	463.1	6,416	1,308.4	18,256
Greedy oracle	24,367.6	6,416	70,376.0	18,256
Learned selector	102.8	6,416	259.9	18,256

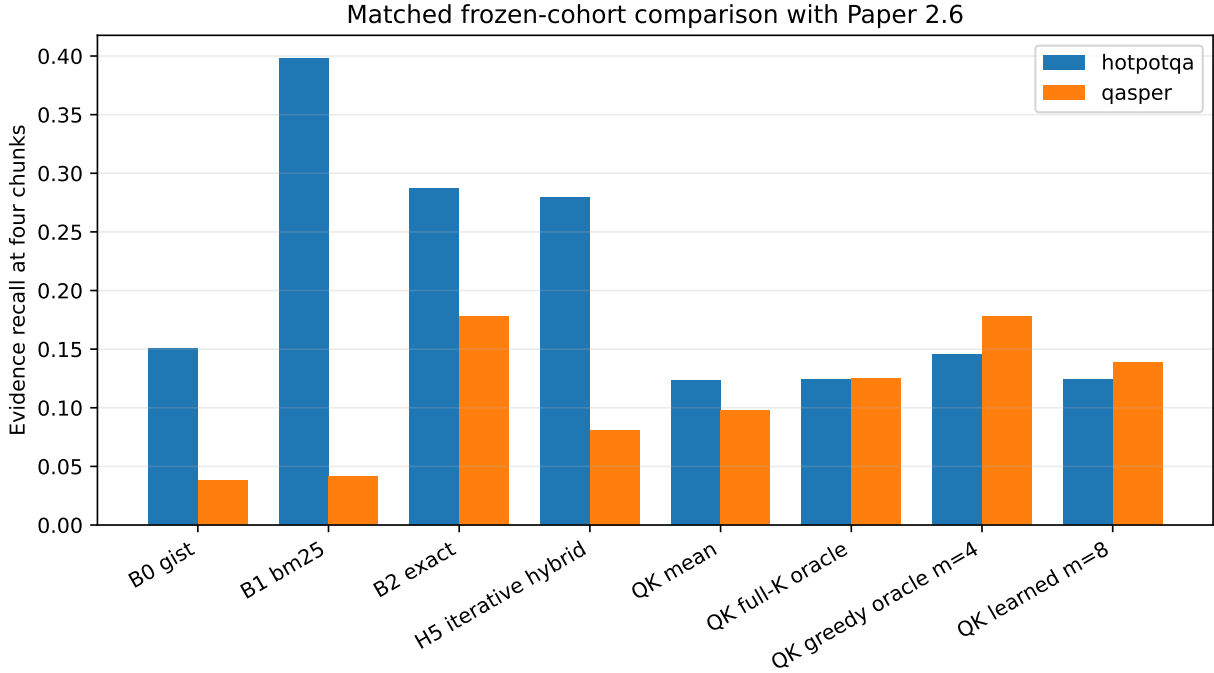


Figure 9: Matched Paper 2.6 comparison. Full-K and greedy rows are non-deployable upper bounds. Learned QK is competitive on QASPER but not Hotpot.

Paper 2.5’s native-QK closure used 256-token parents and a 20% final-parent budget. At that setting, its top-four native closure reached chain completion 0.125 on HotpotQA and 0.875 on QASPER, compared with one-shot parent values 0.138 and 0.938. Those rows concern associative successor discovery, not one-shot 32-token ranking. We preserve them as historical controls but do not pool their metrics with Table 1.

4.9 Cost

The synchronized CUDA audit includes uncached controller construction, compact-key gathering, and QK scoring. Table 7 reports means over all 32 test identities on an NVIDIA GTX 950M. Absolute times are hardware-specific; the ordering and native-dot counts expose algorithmic differences.

The learned selector is 4.5–5.0 times faster than farthest-first and about 237–271 times faster than the top- r greedy oracle, but 16.9–24.4 times slower than mean on this unoptimized Python/CUDA path. Greedy cost is dominated by query-aware candidate search, not compressed scoring. The quality CSV’s `compression_ms` field times compact assembly after indices are cached; Table 7 instead uses the

separate synchronized uncached audit.

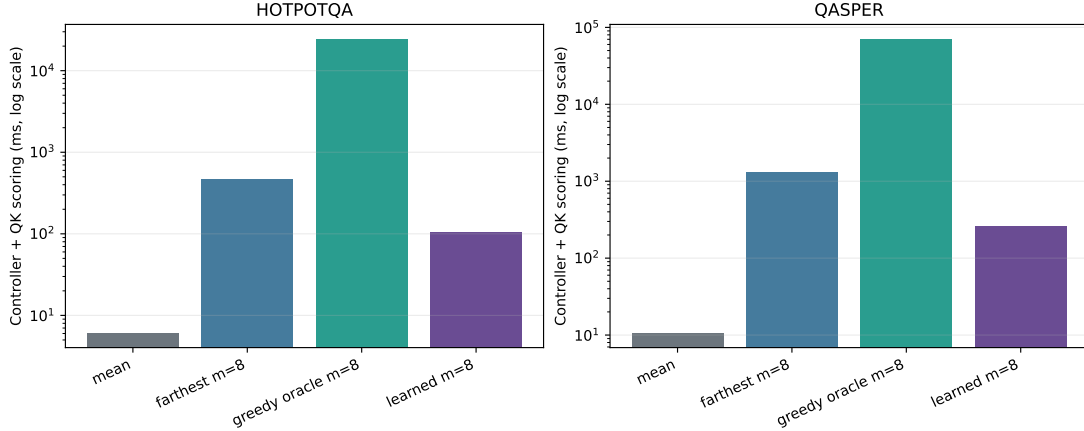


Figure 10: Controller construction plus QK scoring on a log scale. The oracle is a feasibility instrument, not a serving method.

4.10 Gate summary

Table 8: Prespecified decisions. The sequence stops at G3.

Gate	Status	Evidence
G0	Pass	224 inherited Paper 2.6 rows reproduce with zero aggregate mismatches.
G1	Pass	Controlled recall 1.00 vs. 0.95; full-K improves QASPER mean-key recall by 0.0276.
G2	Pass	Greedy $m = 8$ top-four overlap +0.3438; greedy $m = 4$ QASPER recall +0.0802.
G3	Fail	Key-only recovers 40.5%; the best query-conditioned oracle-imitation row recovers 47.3%. Both are below 80%.
G4	Closed	Synthetic off-manifold slots require G3.
G5	Closed	Streaming recurrent memory requires an offline compressor pass.

Table 9: Post-G3 diagnostic decisions. These extension gates do not reopen or revise G3.

Gate	Status	Evidence
E0	Pass	Frozen rows, hashes, model revision, feature caches, and identity partitions reproduce.
E1	Pass	Query/response-aware selectors are not explained by matched native geometry alone.
E2	Inconclusive	The prespecified query-conditioned controller improves Hotpot but regresses QASPER; a test-selected row cannot pass a confirmatory gate.
E3	Exploratory pass	Decision-aware training exceeds oracle imitation by 0.0300/0.0488 recall on Hotpot/QASPER.
E4	Pass	Direct low-rank routing retains the measured retrieval gain with at least $32\times$ less state than the full token-key stream.
E5	Pass	Rank-8, $m = 8$ static compression exceeds the old QASPER semantic gist at 256 bytes per chunk.
E6	Reopened separately	A later frozen 64+64 confirmation validates the QASPER-over-mean effect and permits the reported SmolLM portability test; it does not retroactively pass G3.

5 What the result means

Four observations explain the outcome.

First, a full-key teacher is not automatically an evidence oracle. Native attention keys encode many functions besides document retrieval, and a question’s final layer-27 query can respond to syntactic, discourse, and answer-format features. Top-four mean is the best validated reduction here, but its natural advantage over a mean key is small.

Second, response approximation and top-four selection are different objectives. Greedy $m = 8$ nearly reproduces the teacher ranking on HotpotQA yet does not improve recall. On QASPER, $m = 4$ retrieves more evidence than $m = 8$ despite lower teacher overlap. Additional landmarks can faithfully restore teacher preferences that are irrelevant or harmful to evidence selection. A future objective should combine teacher distillation with validation-only ranking calibration rather than treating the teacher as ground truth.

Third, query conditioning addresses the key-only selector’s missing information need, but does not remove objective mismatch. The combined loss gives the best HotpotQA result and a poor QASPER result; decision-aware training gives the best QASPER result; oracle imitation gives the best response preservation. Capacity alone is not the explanation: increasing rank and landmark count is not monotone, and a controller can reproduce the teacher while retrieving less evidence.

Fourth, direct projection changes the problem. Instead of choosing a few full-width keys and asking them to imitate a native response, it learns a compact interaction space in which evidence chunks rank highly. The QASPER gain survives a 64-fold reduction from the full native-key stream at rank 16, and an eight-centroid rank-8 index preserves a smaller gain at 256 bytes per chunk. Yet low teacher overlap and the Hotpot failure mean this is task-adapted routing, not general native-response distillation. The result supports low-rank native features as one channel inside a selector, not as a replacement for exact, lexical, or hybrid discovery.

The confirmation changes the next question. More selector search is not the priority: QASPER retrieval transfers across cohorts and model families, while generation controls identify frozen memory integration as the limiting stage. The next experiment should keep the selector fixed and train or

calibrate bounded multi-layer memory consumption, with shuffled and bottom-ranked controls retained throughout. Synthetic slots and recurrent state remain premature because neither improves the demonstrated consumer bottleneck.

6 Relation to prior PRA results

Paper 1 established mean-gist routing and selected native K/V. Paper 2.5 tested iterative associative closure and showed that local/native propagation can solve controlled bridges without yielding a uniform natural gain. Paper 2.6 demonstrated strong channel asymmetry: lexical evidence is particularly effective for Hotpot and exact matching for QASPER. Paper 2.7 moved structure discovery to the query side and found that better natural facet recovery could still reduce retrieval.

Paper 2.8 does not rename those mechanisms. It changes the memory-summary objective. Its primary HotpotQA improvement does not overturn Paper 2.6: lexical relational discovery still dominates a native response summary. Its exploratory QASPER landmark result adds something narrower: decision-aware selection of eight native keys can approach, but does not resolve an advantage over, exact matching under the same four-chunk budget. The post-G3 rank-16 router improves further on QASPER while using half the routing state of one full-width mean key, but it is evidence-supervised and its interval against exact routing includes zero. Query-graph facets remain a query-side control and are not used to train the memory selector.

7 Limitations

The fresh confirmation has 128 identities, but only two datasets, two small frozen models, one late routing layer per model, one 32-token chunk size, and one four-chunk budget. Five seeds quantify selector initialization, while identity bootstrap intervals quantify cohort variation; neither establishes broader domain generalization. The original 24-configuration and post-G3 matrices remain exploratory even though the frozen rank-16/rank-8 choices now have a separate confirmation. Hotpot and QASPER do not cover 2Wiki/MuSiQue, and the controlled suite contains only 20 schematic cases.

The teacher does not include post-RoPE distance, value utility, or layer combinations. Greedy oracle landmarks see the evaluation query and cannot be deployed; evidence-supervised projections use validation labels and are task-adapted rather than unsupervised compression. Native-K/V generation uses a single frozen late layer and 12 greedy tokens; its negative QASPER result does not test trained memory integration, multiple consumption layers, or larger instruction models. Direct low-rank indexes reduce routing state, not backing native K/V. Runtime uses eager PyTorch on an old mobile GPU; INT8 is dequantized for scoring rather than using a fused integer kernel, and centroid construction is an offline Python reference path.

8 Conclusion

Compact native-key response imitation fails as a universal objective, but evidence-supervised low-rank native interaction yields a reproducible QASPER routing channel. On 64 fresh QASPER identities, rank 16 improves recall over a mean key by 0.141; the gain transfers to a Llama-family model at smaller magnitude. It does not transfer to Hotpot, and exact lexical routing remains stronger on QASPER. Native-K/V controls sharpen the boundary: informed memory is substantially better than shuffled or bottom-ranked memory, yet frozen late-layer integration can still underperform no memory. Routing and reading must therefore be trained and evaluated separately. The vectorized quantized index makes the routing mechanism operational at 137–1,056 bytes per chunk without changing native K/V, but does not resolve memory use. The next justified step is trained or calibrated multi-layer memory

integration on the confirmed QASPER selector, not a claim that low-rank QK routing replaces lexical discovery.

A Reimplementation map

The implementation separates mechanism from experiment policy.

- `src/prahf/qk_compression.py` defines GQA head mapping, full-key teachers, landmark gathering, no-training selectors, the key-only and query-conditioned selectors, four training losses, response metrics, and routing metrics.
- The Paper 2.5 feature precomputer, `precompute_native_qk_features.py`, now supports identity-disjoint validation/test offsets while retaining the historical default.
- `experiments/paper2.8_qk_compression/run_gated_study.py` selects the teacher on validation, runs gates in order, trains five seeds only after G2, and emits row-level artifacts.
- `run_cost_benchmark.py` times uncached controller construction separately from cached quality evaluation; `summarize_results.py` builds matched Paper 2.5/2.6 tables and plots.
- `run_query_conditioned_study.py` executes the 120-controller matrix with durable row-level checkpoints; `summarize_query_conditioned.py` computes paired comparisons, response recovery, stability, and plots.
- `run_low_rank_frontier.py` evaluates native geometry and trains direct native- Q/K projections; `run_selector_ablation.py` isolates salience and bilinear terms; `summarize_low_rank_frontier.py` emits paired effects, seed stability, normalized cost frontiers, plots, and E0–E6 decisions.
- Confirmation selection uses `run_confirmation.py`. It freezes and replays five-seed selections on the offset-24 cohort.
- Generation replay uses `run_confirmation_generation.py`. It maps exact chunk identities to unchanged native K/V and emits causal controls with crash-safe row checkpoints.
- `run_cross_model_replication.py` retrains the same projection contracts on SmolLM2/Llama native geometry; `run_production_index_benchmark.py` measures quantized vectorized serving, index residency, and append cost.
- `LowRankRoutingIndex` in `src/prahf/qk_compression.py` owns only projected routing tokens, masks, and optional INT8 scales. It never owns or rewrites backing native K/V.

The core serving path is:

```

q = native_q_projection(frozen_query_state)      # [Hq, Dh]
K = cached_pre_rope_keys                        # [C, T, Hkv, Dh]
x = key_only_features(K)                       # [C, T, 8]
z = selector(x).masked_fill(~token_mask, -inf)  # [C, T]
landmarks = gather(K, topk(z, m=8))             # [C, 8, Hkv, Dh]
scores = top4_mean(q @ landmarks / sqrt(Dh))    # [C]
chunks = stable_topk(scores, k=4)
materialize_native_kv(chunks)                   # unchanged 32-token K/V

```

The query-conditioned path replaces the fourth line with `z = selector(x, q)`. The oracle substitutes a query-aware greedy search and must not be used online.

The direct low-rank serving path avoids full native-width online dots:

```
Zk = cache(Wk @ flatten_native_keys(K))      # [C, T, r]
zq = Wq @ flatten_native_query(q)             # [r]
scores = chunk_reduce(Zk @ zq / sqrt(r))       # [C]
chunks = stable_topk(scores, k=4)
materialize_native_kv(chunks)                  # unchanged K/V
```

Static joint compression replaces each chunk’s T projected tokens with four or eight deterministic centroids, medoids, or farthest representatives. Only the routing index changes; the selected chunk still contributes all of its original native K/V.

B Artifacts

Tracked results live under `docs/papers/shared/results/paper2.8.qk_compression/`. The directory contains natural and synthetic row CSVs, teacher validation, paired effects, changed-selection audits, matched Paper 2.6 and historical Paper 2.5 controls, selector checkpoints, synchronized cost rows, plots, gate decisions, and manifests. The `query_conditioned/` subtree adds all 120 controller runs. The `low_rank_frontier/` and `selector_ablation/` subtrees add 3,776 frontier rows, 1,280 ablation rows, 15 projection checkpoints, paired identity bootstraps, seed statistics, byte/latency accounting, plots, and extension decisions. The 329 MB validation, 621 MB test, and 8.3 MB prepared feature caches are ignored; manifests provide hashes and regeneration commands. The full test cache hash is `edc589cc...75993ae8`; validation is `87b85b25...7533cf8`.

References

- [1] Simão, J. Progressive Retrieval Attention: Model-Bounded Sparse Native-KV for Long Context and URI-Addressed Memory. Project manuscript, 2026.
- [2] Simão, J. Iterative PRA and Associative Memory Closure. Project manuscript, 2026.
- [3] Simão, J. Hybrid Lexical-Semantic Routing for Progressive Retrieval Attention. Project manuscript, 2026.
- [4] Simão, J. Latent Query Decomposition from Frozen Transformer State Graphs. Project manuscript, 2026.
- [5] Yang, Z. et al. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. EMNLP, 2018.
- [6] Dasigi, P. et al. A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers. NAACL, 2021.